

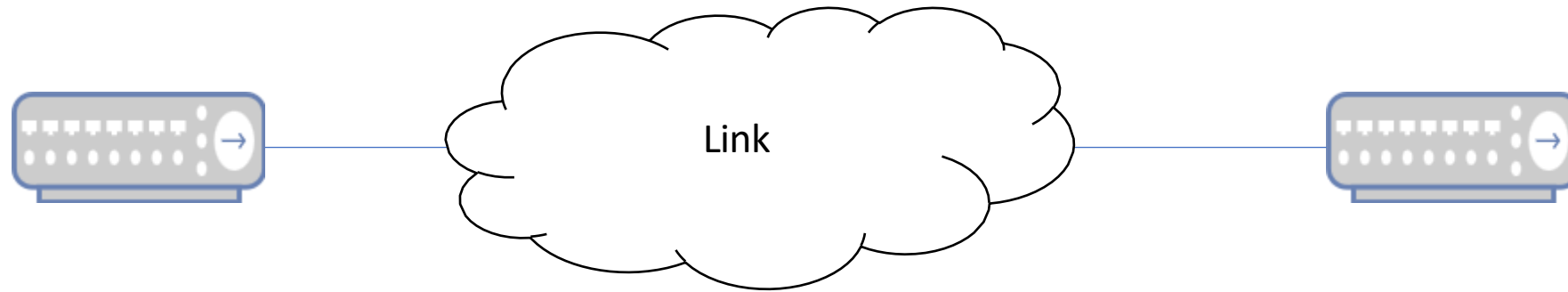
Problem / Overview

Course: Networking Fundamentals
Module: Link



University of Colorado **Boulder**

Introducing a Link

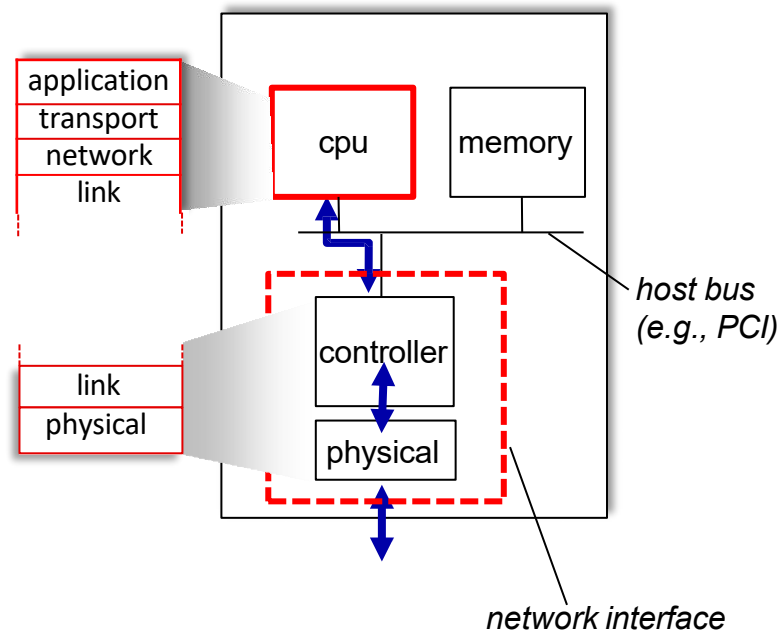


Networks consist of links interconnecting nodes (through wires or wirelessly)



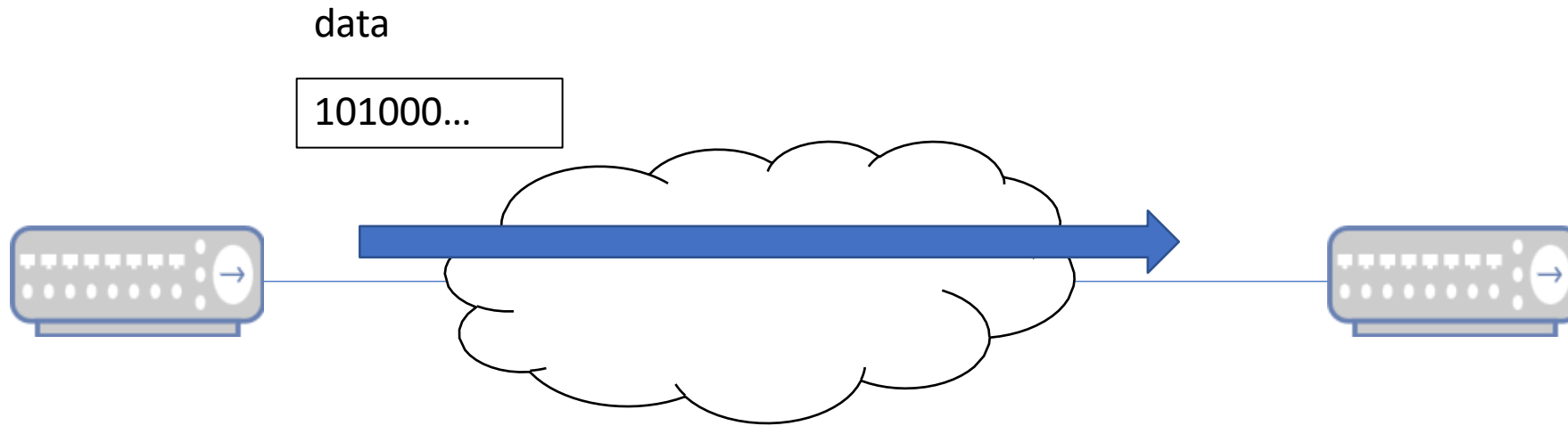
Where is the Link Layer implemented

- Typically, in hardware on network interface card (NIC)



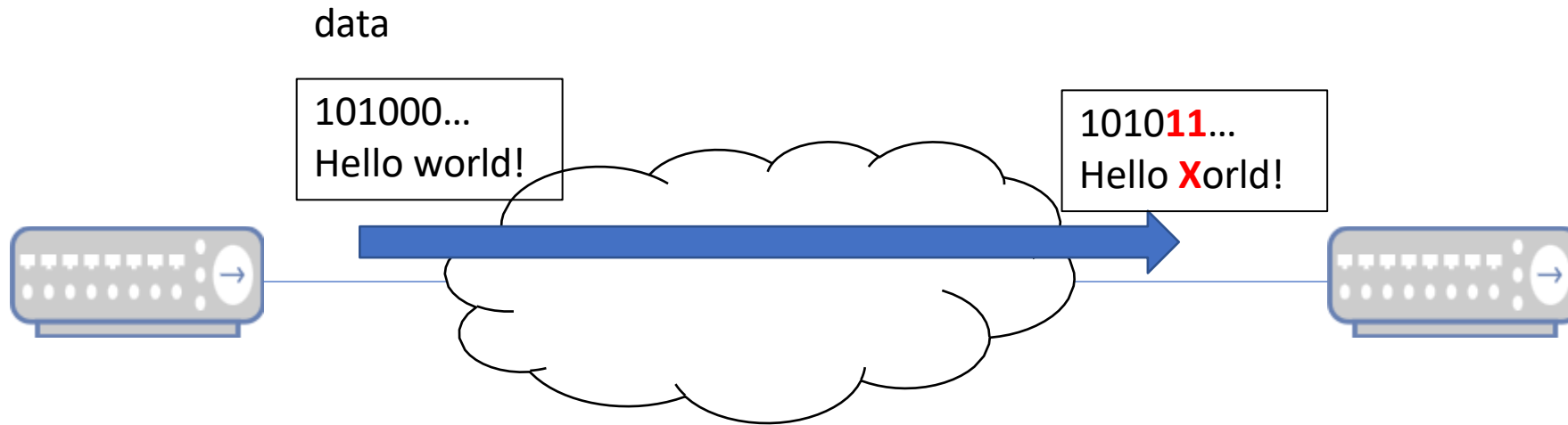
Problems solved by link layer

1. How do we encode the data such that each side can understand who it's destined for, and how to interpret the data.



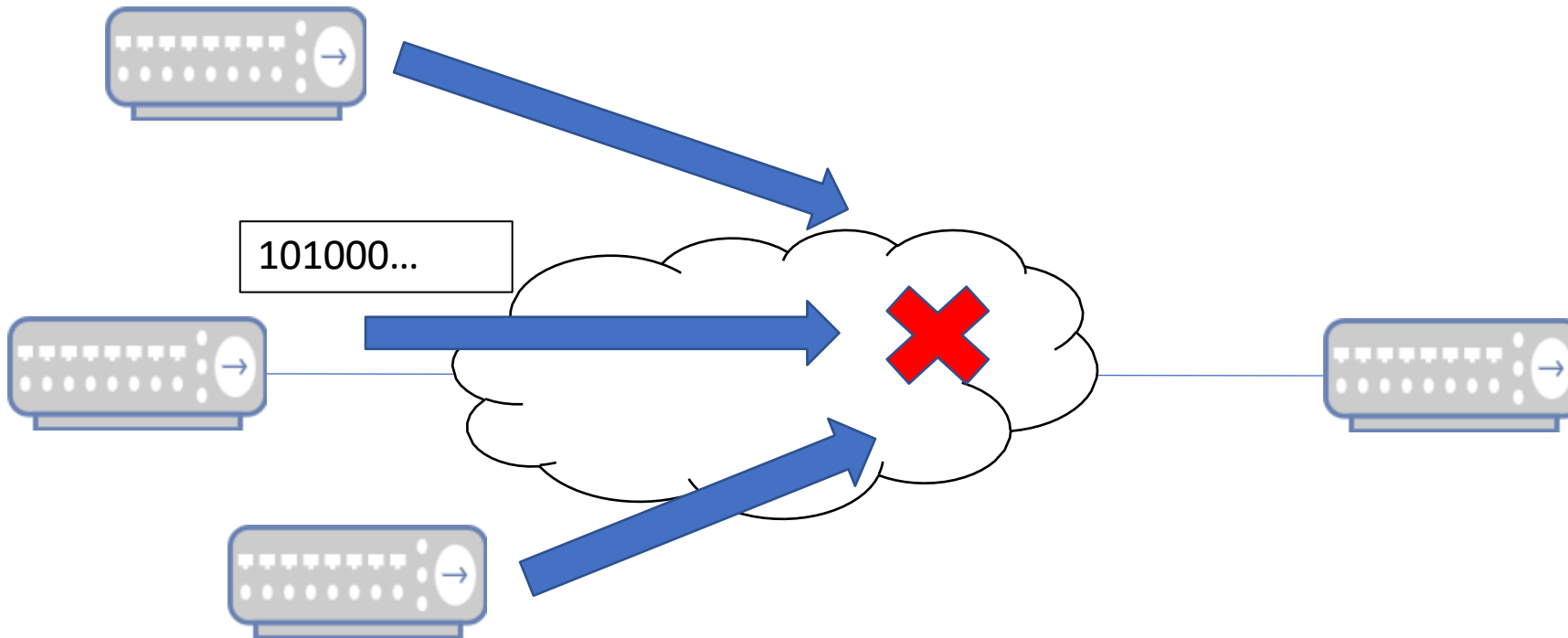
Problems solved by link layer

2. How do we detect if an error has occurred, and better yet, correct it.



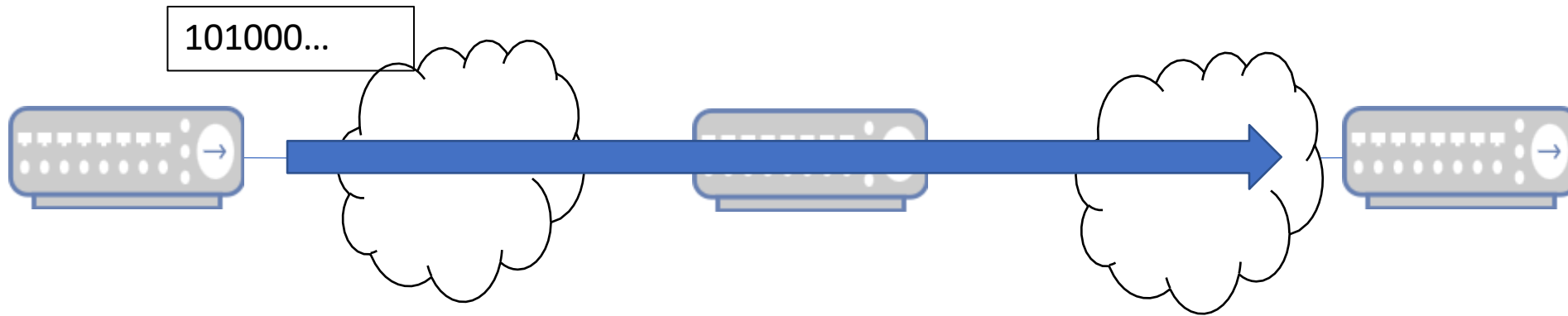
Problems solved by link layer

3. For shared access mediums, how do we coordinate between nodes to regulate access to the medium



Problems solved by link layer

4. How to extend into a local area network (to extend beyond the physical limits of the transmission medium)





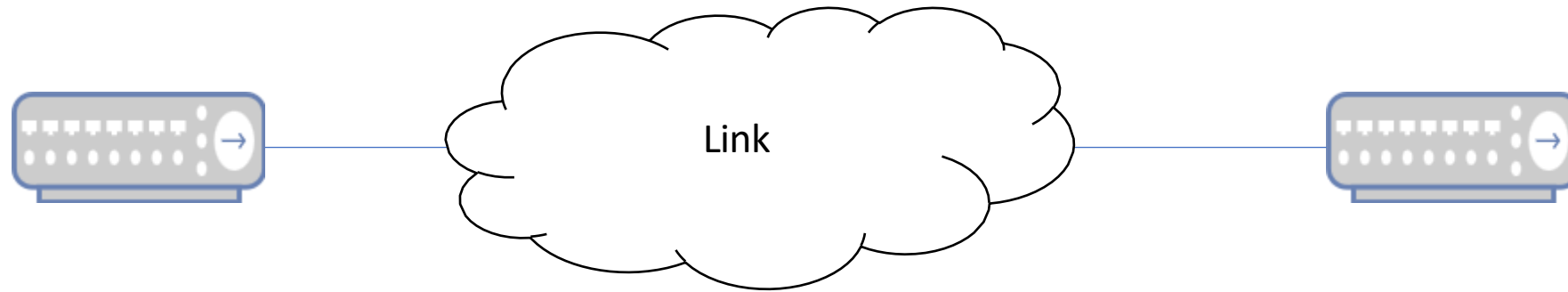
Encoding into Frames

Course: Networking Fundamentals
Module: Link



University of Colorado **Boulder**

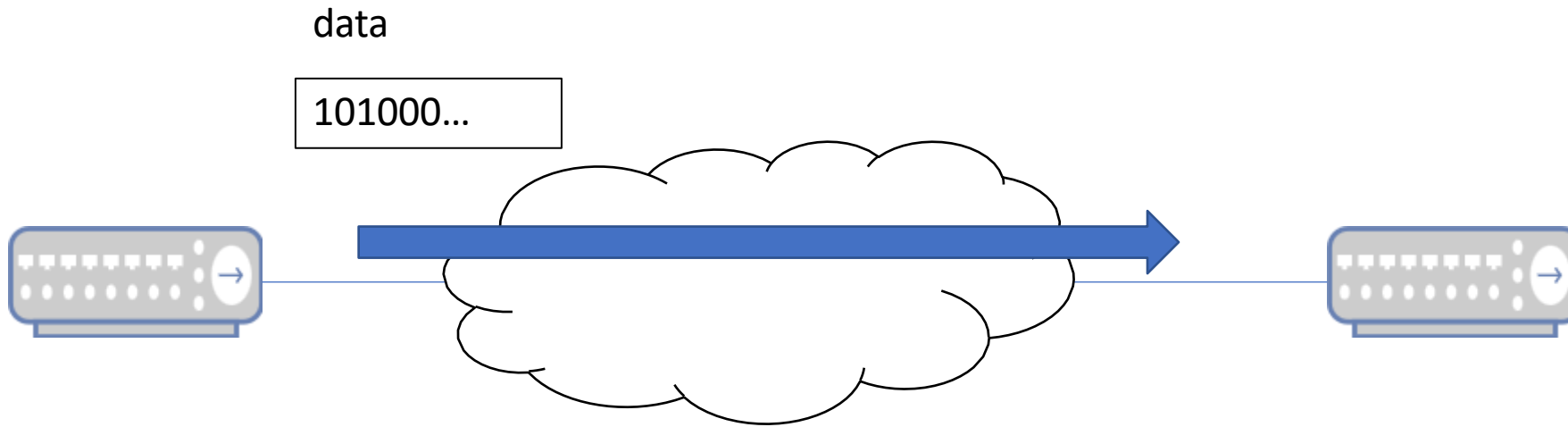
Introducing a Link



Networks consist of links interconnecting nodes (through wires or wirelessly)

Problems solved by link layer

1. How do we encode the data such that each side can understand who it's destined for, and how to interpret the data.



Structuring the data

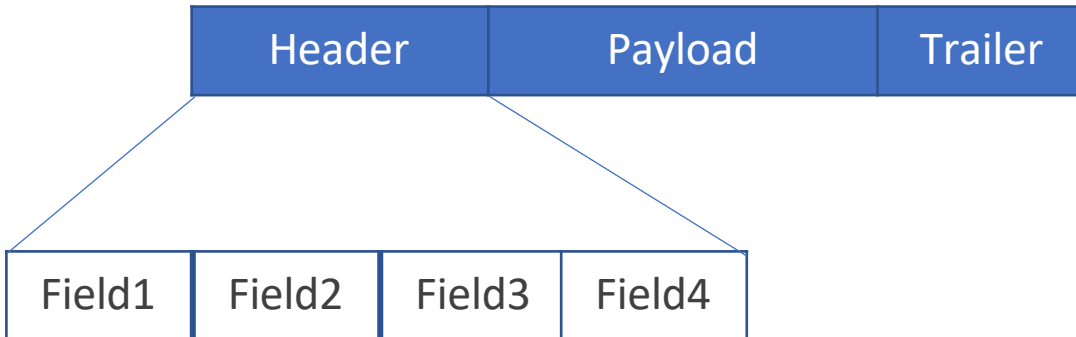
- Assume you can transmit 1s and 0s through whatever medium
- Making sense of the series of 1s and 0s needs structure

➔ Frame



Header / Trailer / Payload

- The payload is the data that is to be transmitted
- The header and trailer are extra information to help understand how to handle it
- Each is broken down into fields



Ethernet

- Dest is for filtering messages (is it for this node)
- Src is useful to know who sent the message (so node can reply)
- Type – tells what the data is
- Frame Check Sequence (FCS) – for error handling

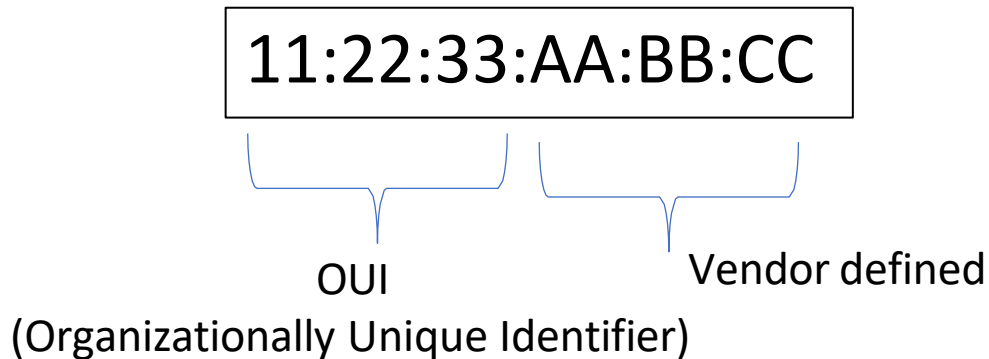
Dest	Source	Type	data	FCS
48 bits	48 bits	16 bits		32 bits

EtherType	Protocol
0x0800	Internet Protocol version 4 (IPv4)
0x0806	Address Resolution Protocol (ARP)
0x0842	Wake-on-LAN ^[8]
0x22F0	Audio Video Transport Protocol (AVTP)
0x22F3	IETF TRILL Protocol
0x22EA	Stream Reservation Protocol
...	



Media Access Control (MAC) address

- 48 bits, commonly displayed in hex with colons
- Historically assigned as part of the manufacturer and globally unique (with virtualization, not the case)



Broadcast

Special destination address which says this is for everybody

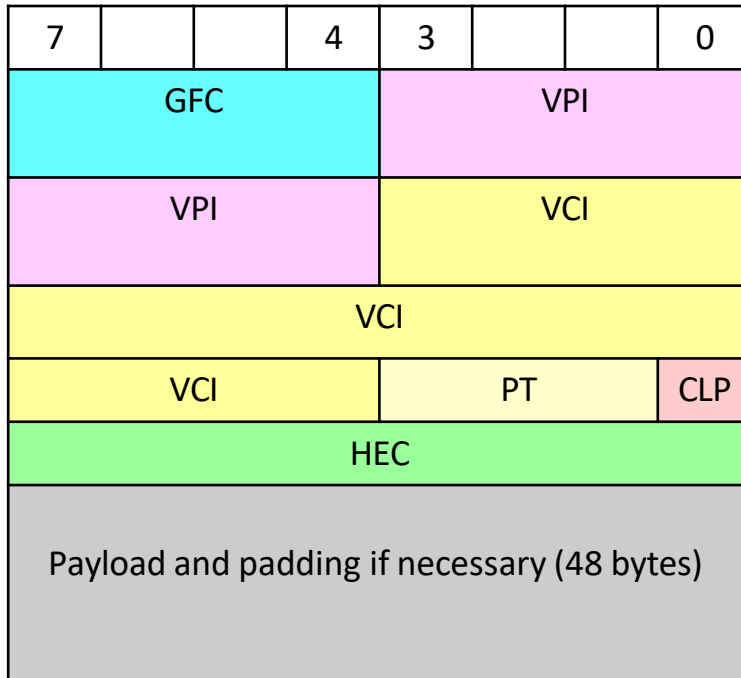
FF:FF:FF:FF:FF:FF



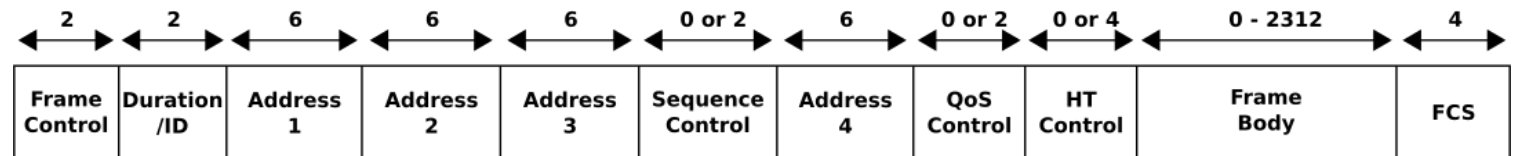
Different Frame Formats

Key take away is that some standard defines the format, so everyone using it can know how to speak to one another

ATM



802.11





University of Colorado **Boulder**

Error Handling

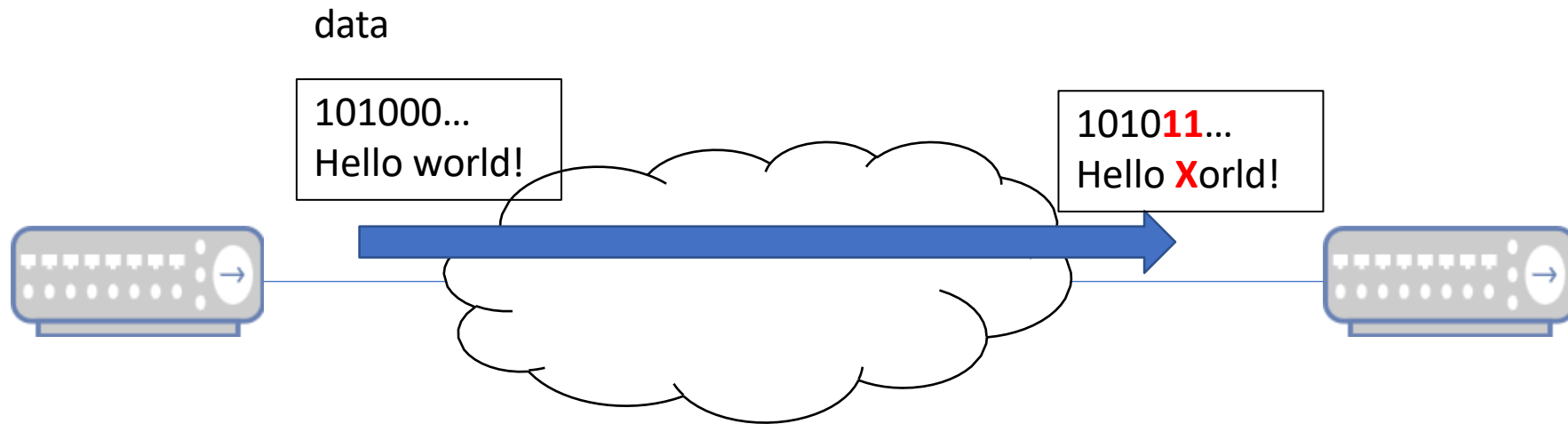
Course: Networking Fundamentals
Module: Link



University of Colorado **Boulder**

Problems solved by link layer

2. How do we detect if an error has occurred, and better yet, correct it.



Detecting an Error With Parity

Send		Receive	
1010 0011	→	1010 1 011	ERROR
1010 0011	→	1010 0011	CORRECT

Goal:

Detect if there was an error in transmission

Parity:

Force every transmission to have an even (or odd) number of 1s by adding 1 extra bit set to 0 or 1 based on number of 1s in data

If a single bit gets flipped, it would make the transmission odd (or even)



Aside: xor

	0	1
0	0	1
1	1	0

$$\begin{array}{r} 0101 \\ \oplus 0011 \\ \hline 0110 \end{array}$$

Counting 1s – use xor

$$1010 \Rightarrow 1 \oplus 0 \oplus 1 \oplus 0 = 0 \text{ (even)}$$

$$1110 \Rightarrow 1 \oplus 1 \oplus 1 \oplus 0 = 1 \text{ (odd)}$$



Parity Example

Even parity (make transmission have an even number of 1s)

Data: 11000000 $\Rightarrow 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 = 0$

(it's already even, so add a 0 to the transmission)

Transmit: 110000000

Data: 01010111 $\Rightarrow 0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 = 1$

(it's odd, so add a 1 to the transmission)

Transmit: 010101111



Parity Example

Received: 110000000

$$1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 \oplus 0 = 0 \quad \text{CORRECT}$$

Received: 010101111

$$0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus 1 = 0 \quad \text{CORRECT}$$

Received: 0101011**0**1

$$0 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 1 \oplus \mathbf{0} \oplus 1 = 1 \quad \text{ERROR}$$



Cyclic redundancy check (CRC)

CRC is a more powerful form of error detection

Cyclic codes are particularly suited to detect burst errors

=> Adopted for the FCS in Ethernet

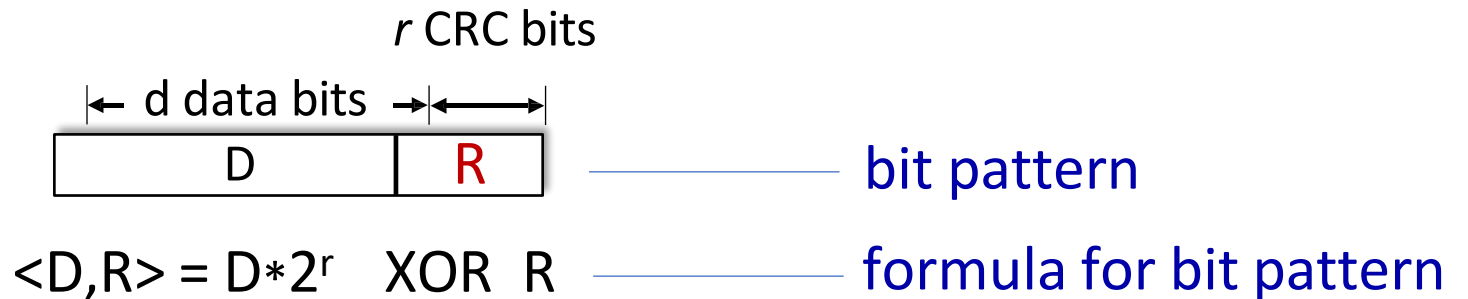


CRC Calculation

Given:

D: Data to be transmitted

G: generator polynomial (value defined by Ethernet standard) ($r+1$ bits)



Calculate: Find R (where $\langle D, R \rangle$ is what will be transmitted) such that:

$\langle D, R \rangle$ is divisible by G (mod 2)

Check: Receiver divides $\langle D, R \rangle$ by G

If remainder is 0, no error. Otherwise, error



Example CRC calculation

- $r = 3$
- $G = 1010$ ($r+1=4$ bits)
- $D = 11101010$
- Find: R (3 bits)



Step 1: Set $R=0$

11101010000
└───┬───┘ └─┘
D R

Setting $R = 0$ allows us to determine what we need to transmit to make $\langle D, R \rangle$ exactly divisible by G



Step 2: Divide by G

$$\begin{array}{r} 1 \\ 1010 \overline{) 11101010000} \\ \underline{1010} \\ 100 \end{array}$$

Quotient bit = 1 if divisor can “fit” (by # of bits),
otherwise 0

Modulo 2 Division simply involves XOR (not subtraction)



Step 2: Divide by G

$$\begin{array}{r} 11 \\ 1010 \overline{) 11101010000} \\ \underline{1010} \\ 1001 \\ \underline{1010} \\ 11 \end{array}$$

1010 can fit in 1001, so quotient bit is 1



Step 2: Divide by G

$$\begin{array}{r} 110 \\ 1010 \overline{) 111010100000} \\ \underline{1010} \\ 1001 \\ \underline{1010} \\ 110 \end{array}$$

1010 can't fit in 100, so quotient bit is 0

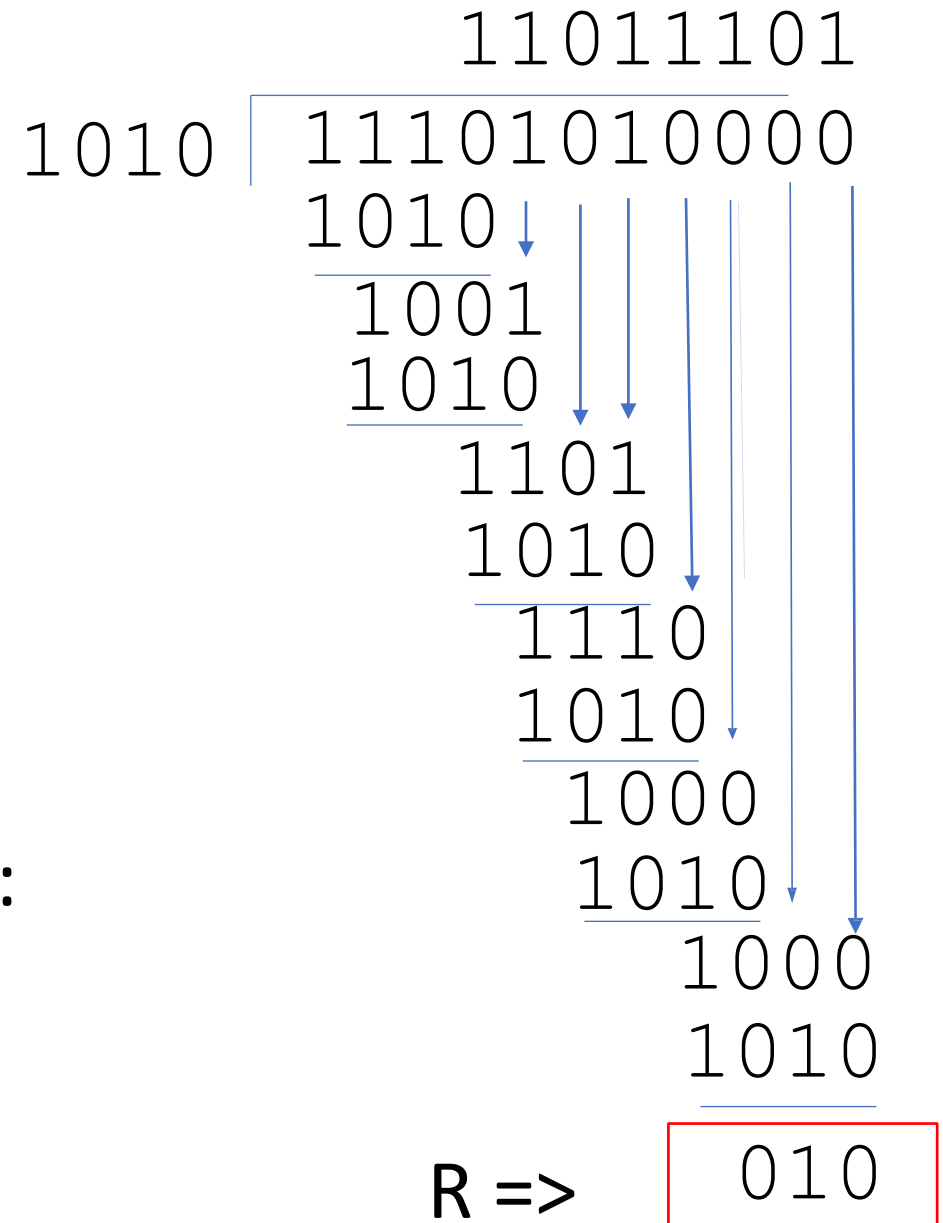


Step 2: Divide by G

R is 010, so would transmit $\langle D, R \rangle$:

11101010010

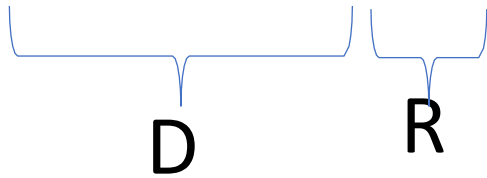
111010100 10
D R



Step 3: Transmit

R is 010, so would transmit $\langle D, R \rangle$:

11101010010



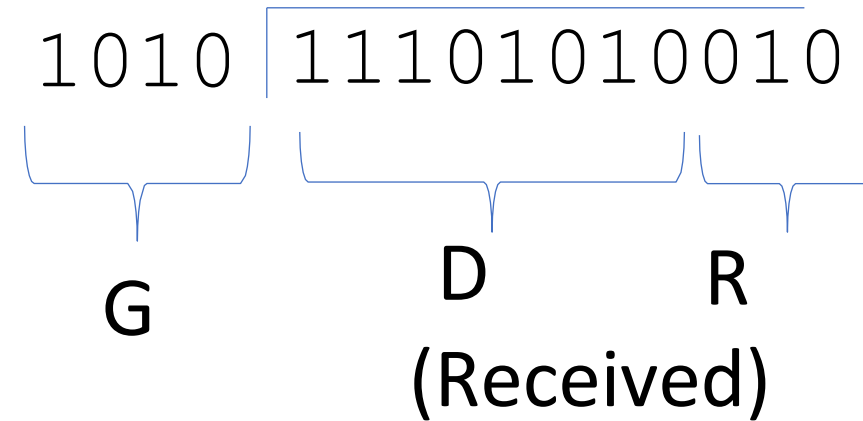
Check on Receive Side

Received $\langle D, R \rangle$

Divide by G

If Remainder is 0 $\Rightarrow$ correct

Else $\Rightarrow$ Error



CRC in Ethernet

Detects:

- Any 1 bit error
- Any two adjacent 1 bit errors
- Any odd number of 1 bit errors
- Any burst of errors with a length of 32 or less

Can be calculated as bytes arrive off of the wire

- Simple calculation only involving xor operation



Ethernet FCS

3.2.9 Frame Check Sequence (FCS) field

A cyclic redundancy check (CRC) is used by the transmit and receive algorithms to generate a CRC value for the FCS field. The FCS field contains a 4-octet (32-bit) CRC value. This value is computed as a function of the contents of the protected fields of the MAC frame: the Destination Address, Source Address, Length/Type field, MAC Client Data, and Pad (that is, all fields except FCS). The encoding is defined by the following generating polynomial.

$G = 04C11DB7$

$$G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$$

Mathematically, the CRC value corresponding to a given MAC frame is defined by the following procedure:

- The first 32 bits of the frame are complemented.
- The n bits of the protected fields are then considered to be the coefficients of a polynomial $M(x)$ of degree $n - 1$. (The first bit of the Destination Address field corresponds to the $x^{(n-1)}$ term and the last bit of the MAC Client Data field (or Pad field if present) corresponds to the x^0 term.)
- $M(x)$ is multiplied by x^{32} and divided by $G(x)$, producing a remainder $R(x)$ of degree ≤ 31 .
- The coefficients of $R(x)$ are considered to be a 32-bit sequence.
- The bit sequence is complemented and the result is the CRC.

Implementation
to deal with case
of leading 0s

D is frame data,
M is $\langle D, R \rangle$
Divide by G and
output is R



For further explanation and implementation

Understanding CRC32 by Joshua Davies

<https://commandlinefanatic.com/cgi-bin/showarticle.cgi?article=art008>





Sharing the Link

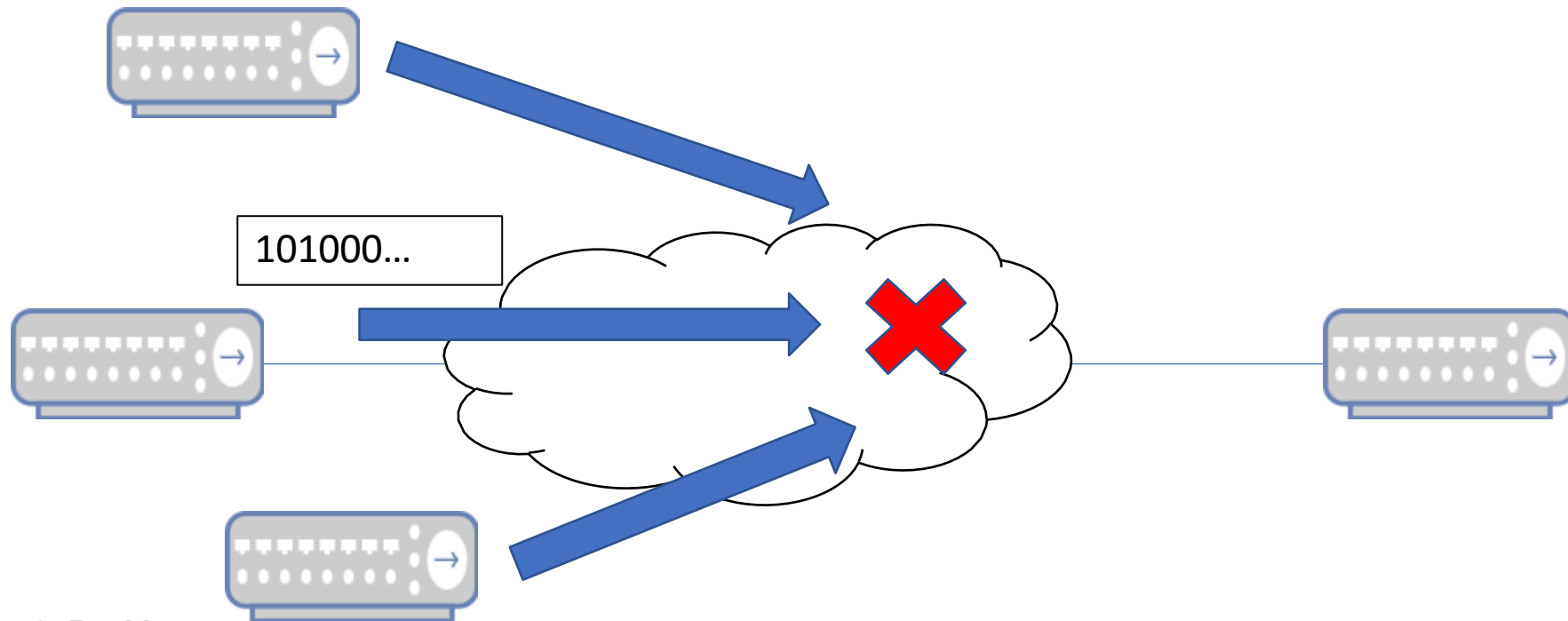
Course: Networking Fundamentals
Module: Link



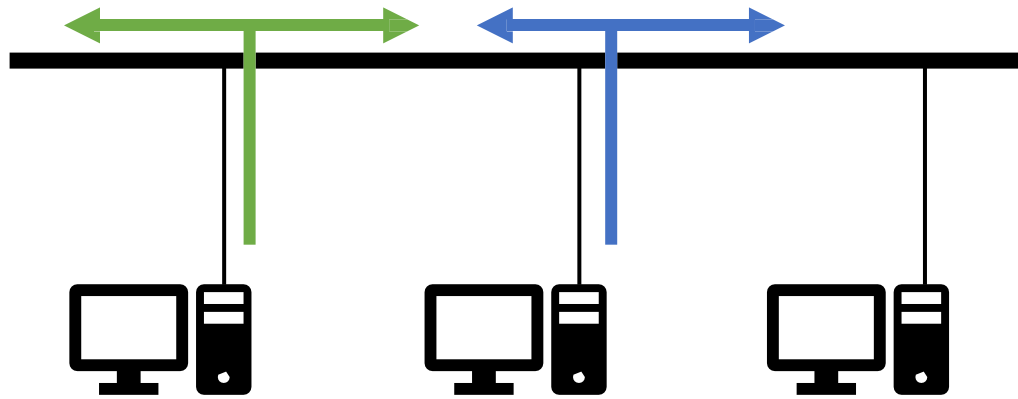
University of Colorado **Boulder**

Problems solved by link layer

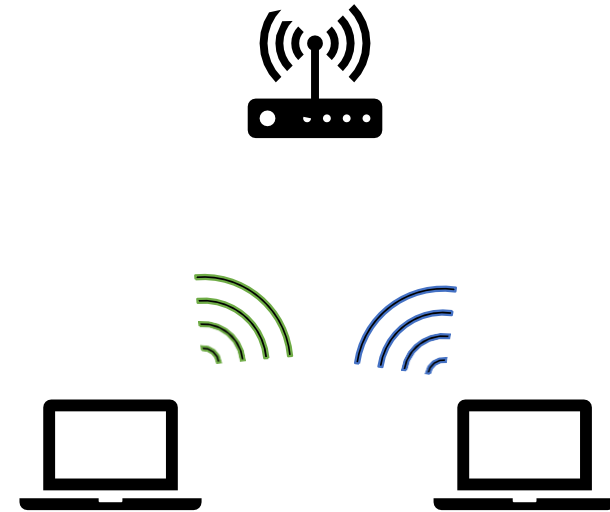
3. For shared access mediums, how do we coordinate between nodes to regulate access to the medium



Interference



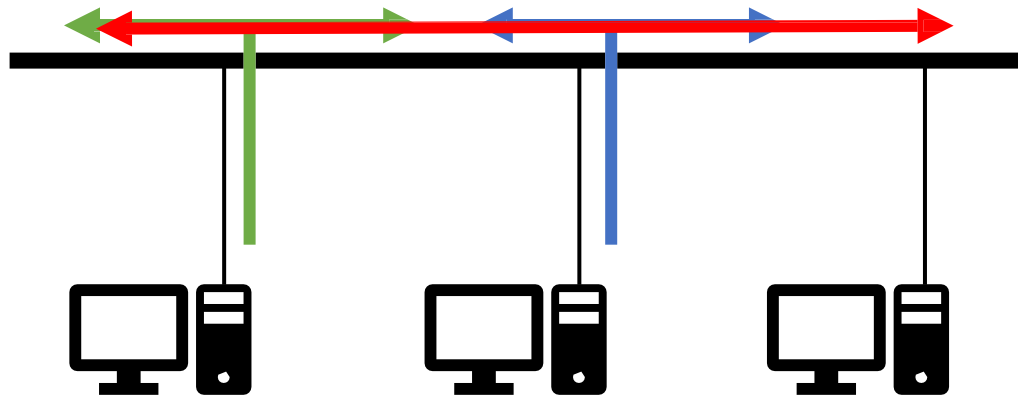
Wired



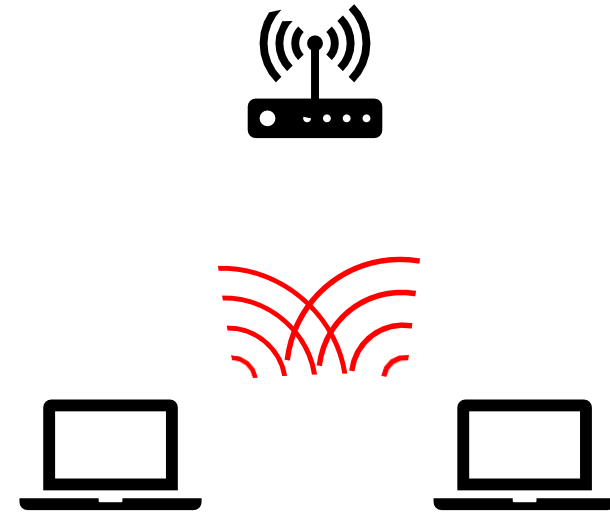
Wireless

Interference

When collisions happen – no data is usefully transmitted



Wired



Wireless

Multiple Access Protocols

How do we share the link fairly and efficiently?



Possible Approaches

Channel Partitioning:

- Divide the link into smaller pieces and assign to each node

Random

- Nodes can transmit in an uncoordinated manner
- Deal with collisions when they happen



Channel Partitioning - Time

Time Division Multiple Access (TDMA)

- Divide into equal sized time slots



- Assign slots to nodes

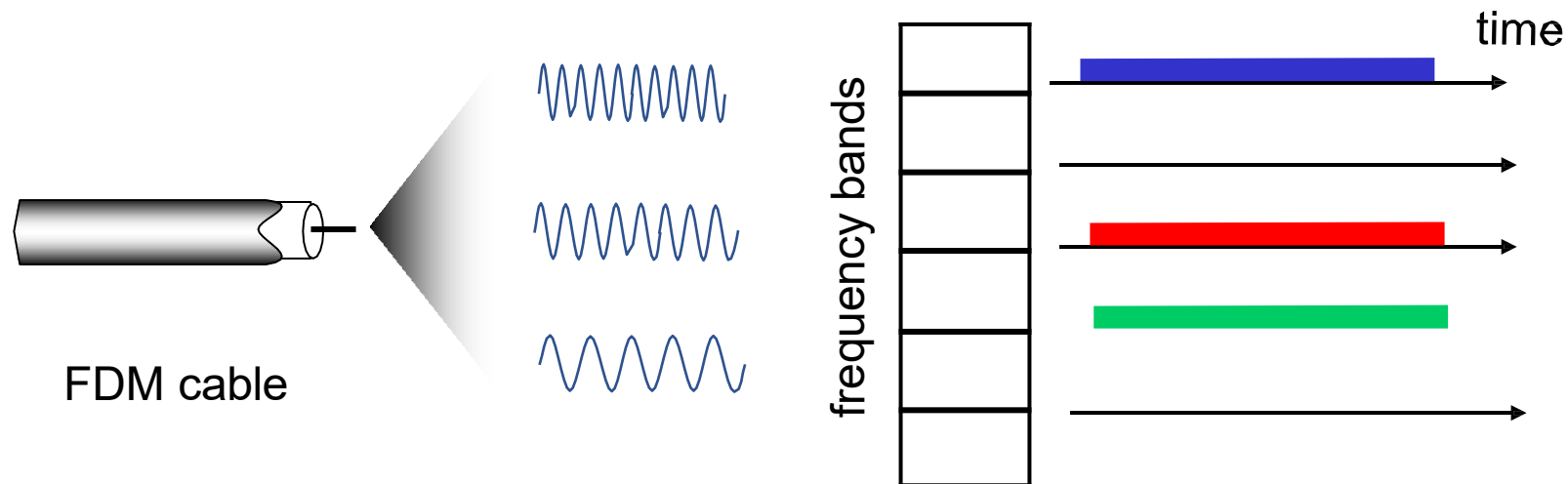


- Nodes can send during their slots only
 - Ex: Slots are 100 bytes. 1 has 10 bytes to send, 2 has 0, 3 has 200, 4 has 150



Partition Channel - Frequency

- Frequency Division Multiple Access (FDMA)
- Divide frequency spectrum and assign each node it's own frequency



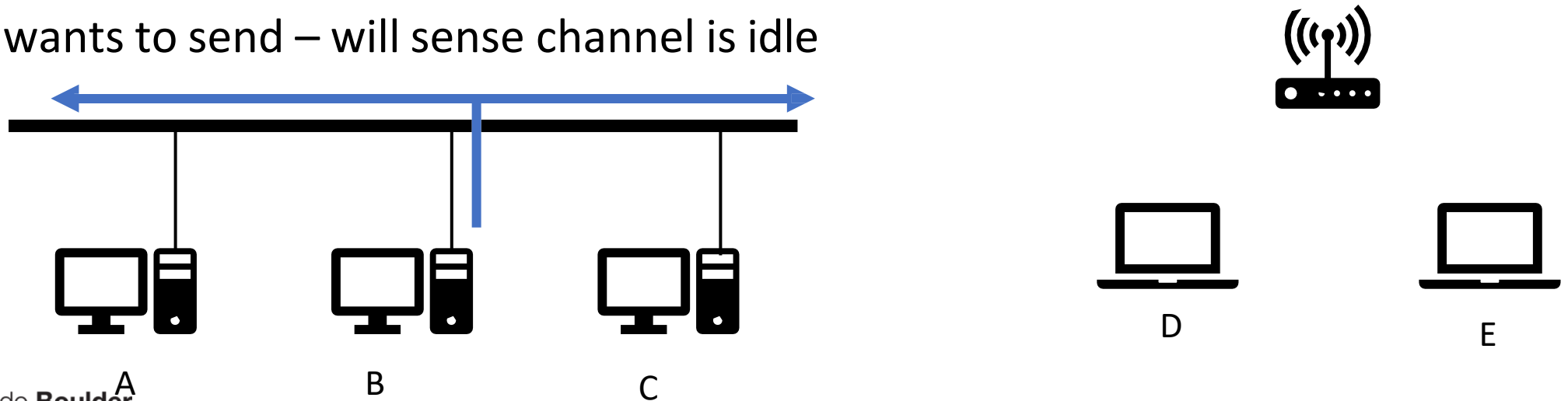
Random Multi Access Protocols

- When to send?
 - Anytime? Make an attempt to at least not interfere?
- How does a node know if transmission was successful?
 - Can a node detect this? Does the receiver have to acknowledge?
- What happens if transmission failed?
 - Resend immediately? Wait for some amount of time?



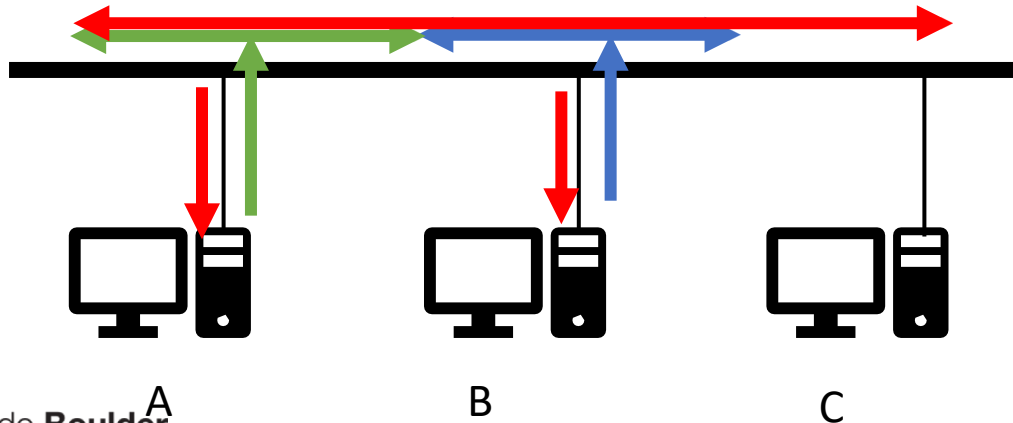
When to Send

- Carrier Sense Multiple Access (CSMA)
 - Used in both Ethernet (802.3) and Wi-Fi (802.11)
- Node will read from the channel to see if another node is sending and transmits when it senses idle
- Example:
 - A wants to send – will sense channel not idle (since B is sending)
 - D wants to send – will sense channel is idle



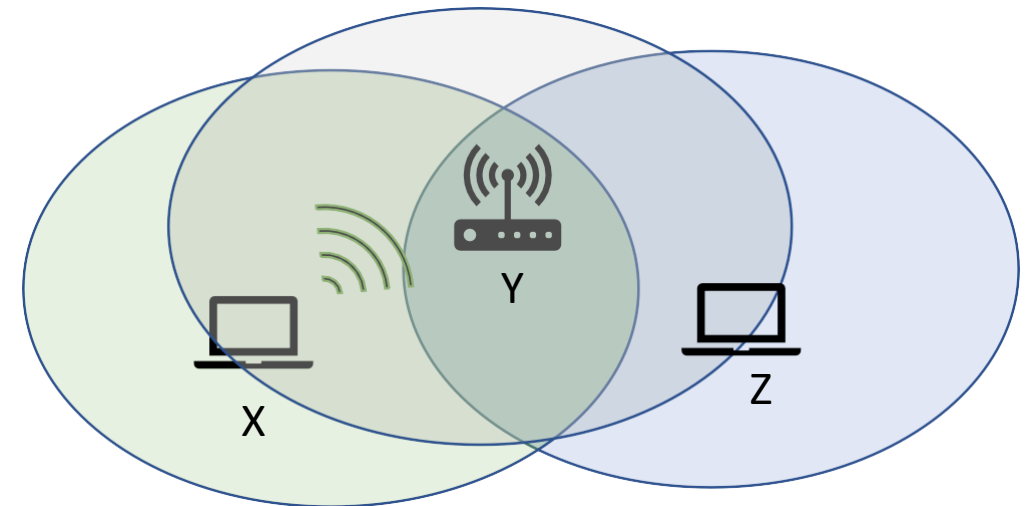
Knowing if a Transmission was Successful (CSMA / CD - Ethernet)

- CSMA / CD – collision detection (802.3)
- Ethernet NICs can read while transmitting
- If what it reads is different than what it's sending, it detects a collision
- Node will send a “jam” signal – transmit fixed bit pattern to ensure everyone detects the collision



Knowing if a Transmission was Successful (CSMA / CA – Wi-Fi)

- CSMA / CA – collision avoidance (802.11)
 - 802.11 NICs cannot read while transmitting and may be out of range of some nodes (Z can't sense X is sending)
 - It depends on an explicit acknowledgement from receiver
 - If it doesn't receive an ACK in certain period of time, assumes collision
-
- Optional RTS / CTS:
 - Request to send
 - Clear to Send



When a Transmission Fails

Re-transmit the frame some time period (backoff)

CSMA/CA and CSMA/CD specify:

- Random backoff time – to avoid nodes repeatedly colliding
- Exponentially increasing – when multiple transmissions fail, wait longer each time





Local Area Network

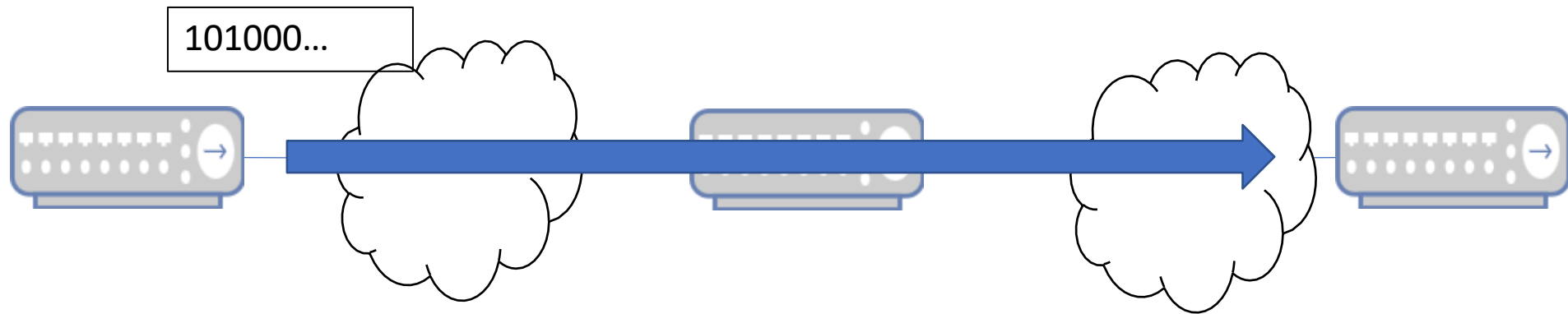
Course: Networking Fundamentals
Module: Link



University of Colorado **Boulder**

Problems solved by link layer

4. How to extend into a local area network (to extend beyond the physical limits of the transmission medium)



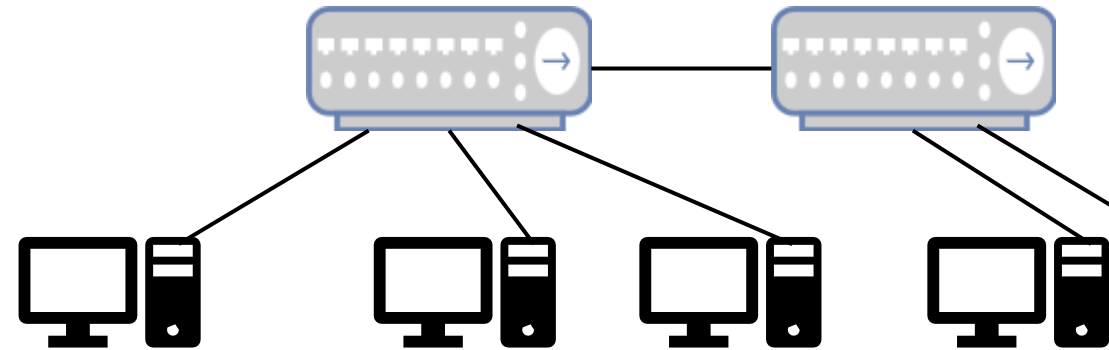
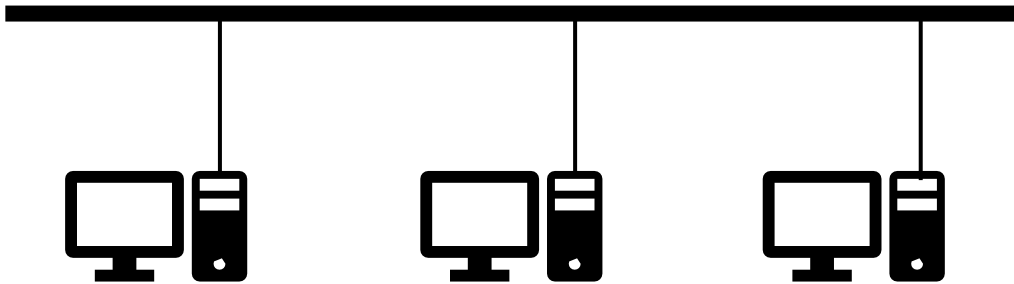
Topics Covered

- Ethernet switching
- Wireless access points



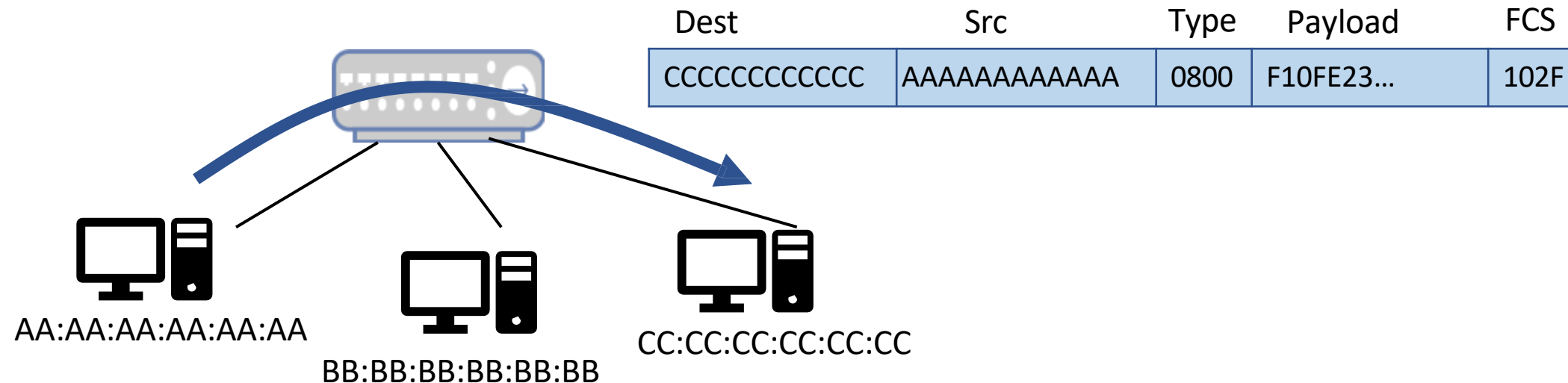
Ethernet Switching

- Scalability challenges with shared cable (left)
- Switching solves this (right)
 - Uses point to point full-duplex links
 - Special nodes (switches) can direct traffic to destination



Ethernet Addressing

- Recall:
 - Each NIC has a MAC address
 - Each frame contains src and dest MAC address

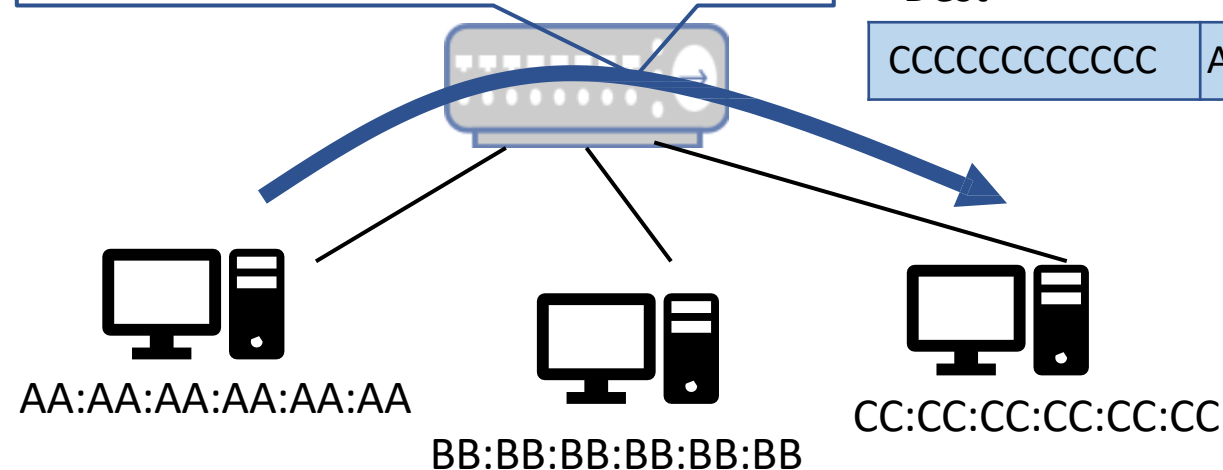


MAC Address Table

- Table in switch maps destination addresses to output ports
- Receive frame, lookup destination, send frame

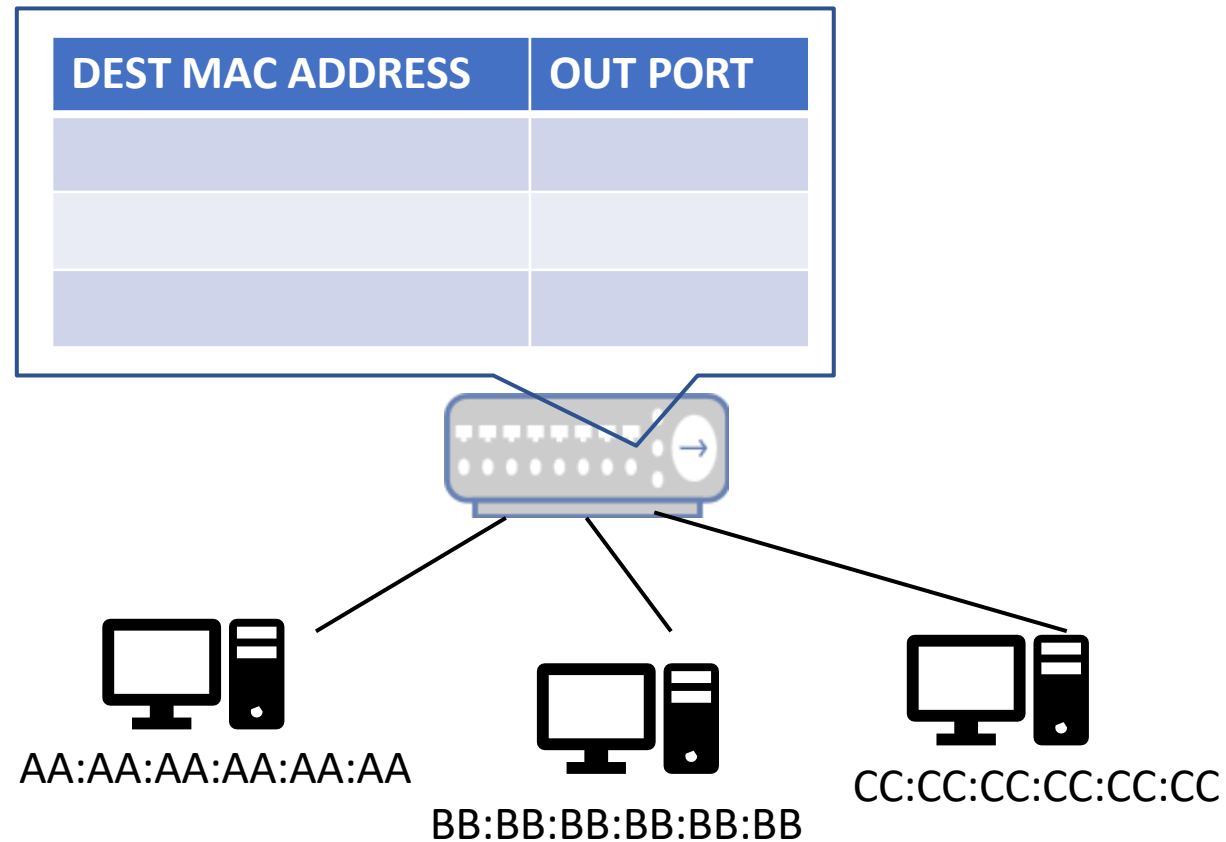
DEST MAC ADDRESS	OUT PORT
AA:AA:AA:AA:AA:AA	0
BB:BB:BB:BB:BB:BB	1
CC:CC:CC:CC:CC:CC	2

Dest	Src	Type	Payload	FCS
CCCCCCCCCCCC	AAAAAAAAAAAA	0800	F10FE23...	102F



How do we Fill the MAC Address Table?

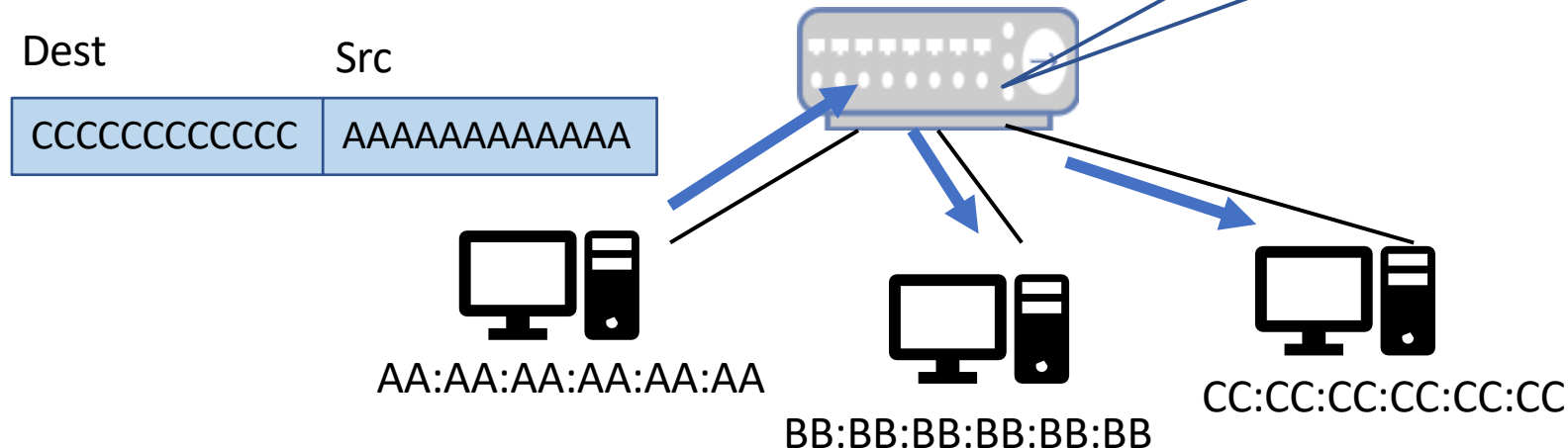
- Table starts empty



MAC Table Learning

- When switch receives a frame it **learns** the source is at that port
- If destination is unknown, **flood** to all ports
 - All but the actual destination will ignore it

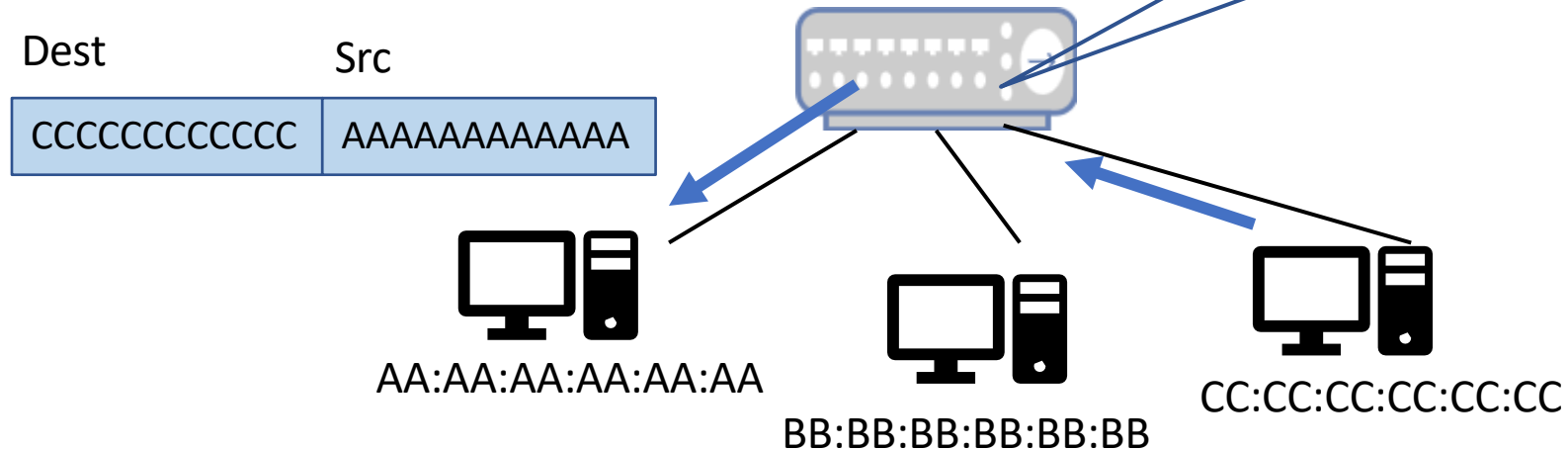
DEST MAC ADDRESS	OUT PORT



MAC Table Learning

- If destination replies, switch will learn the address on that port

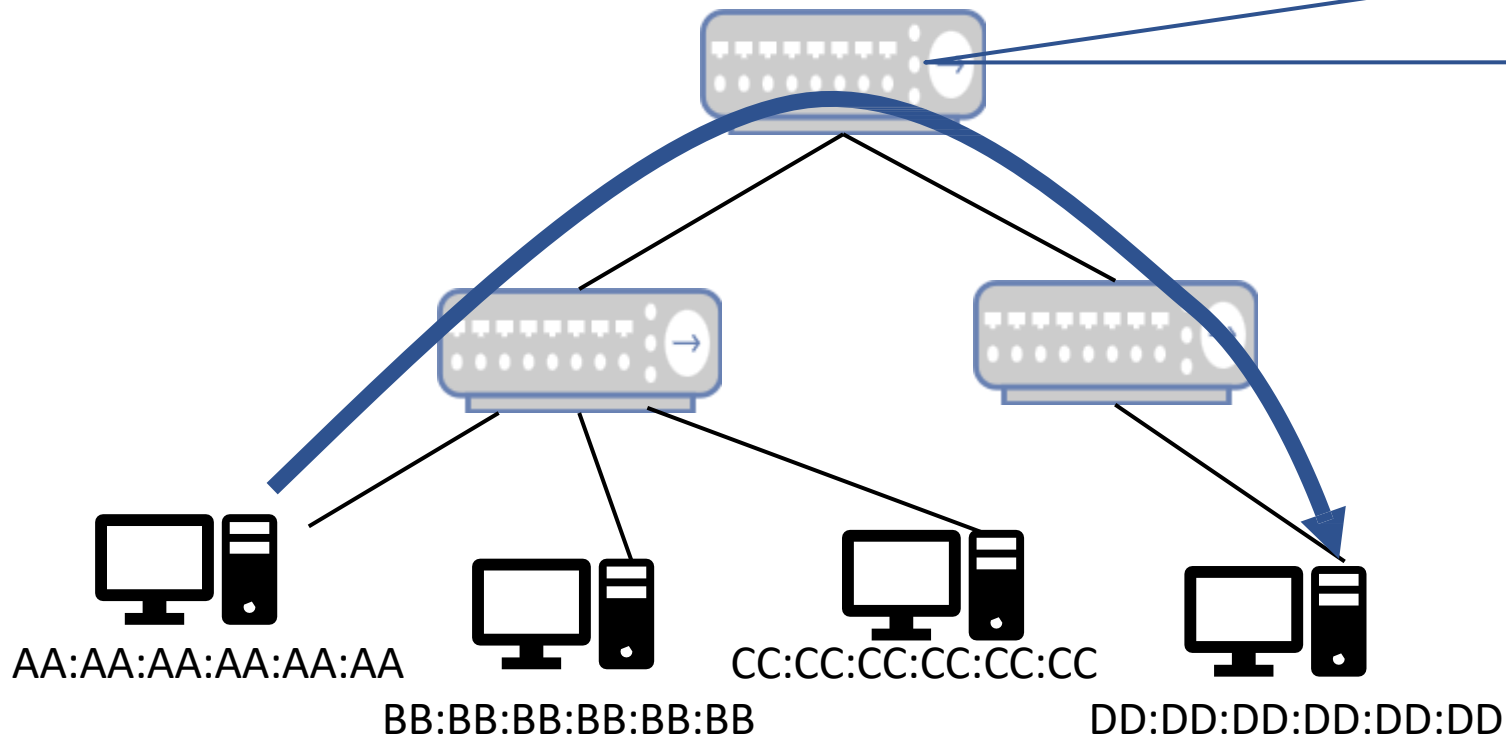
DEST MAC ADDRESS	OUT PORT
AA:AA:AA:AA:AA:AA	0



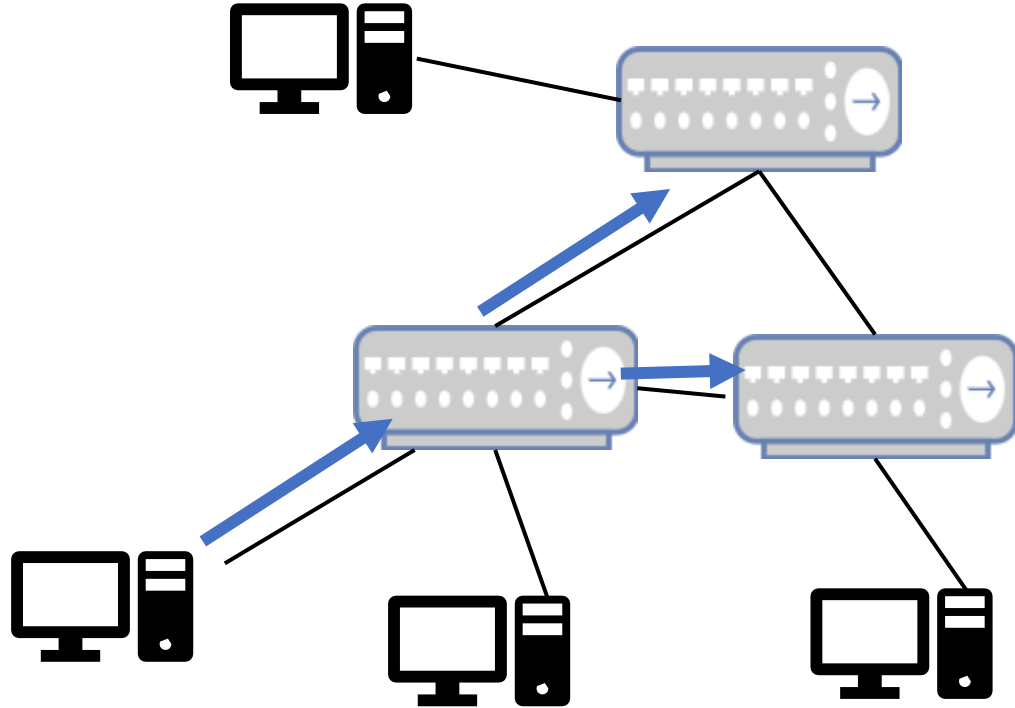
Topology of Switches

- Switches are limited in number of ports, so you can extend with a topology of switches

DEST MAC ADDRESS	OUT PORT
AA:AA:AA:AA:AA:AA	0
BB:BB:BB:BB:BB:BB	0
CC:CC:CC:CC:CC:CC	0
DD:DD:DD:DD:DD:DD	1

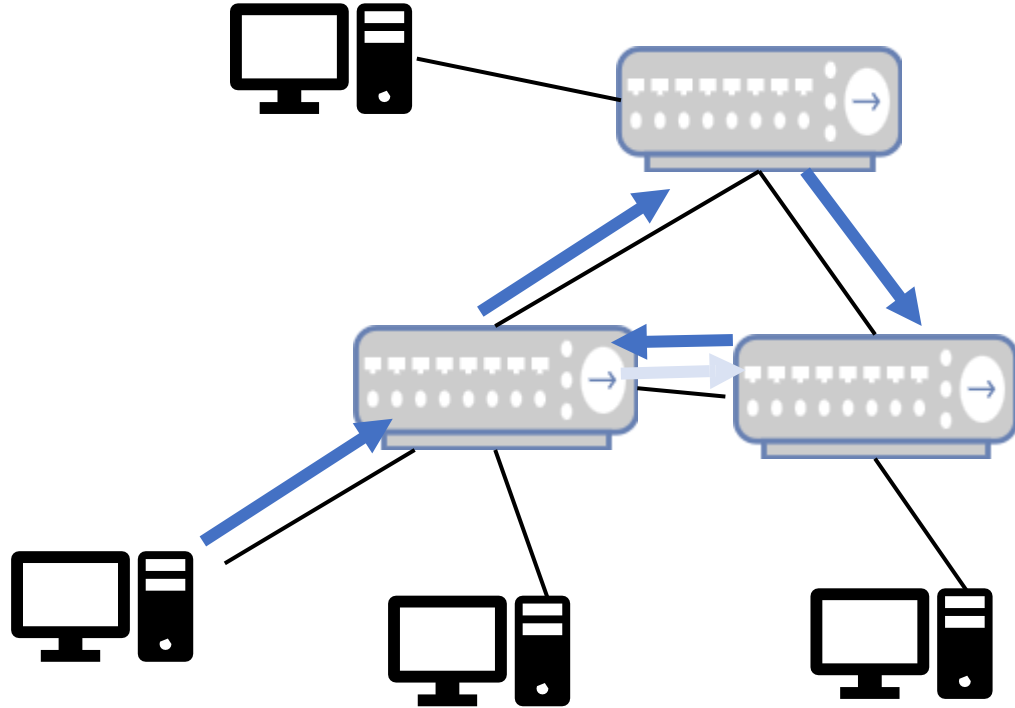


Problem: Loops



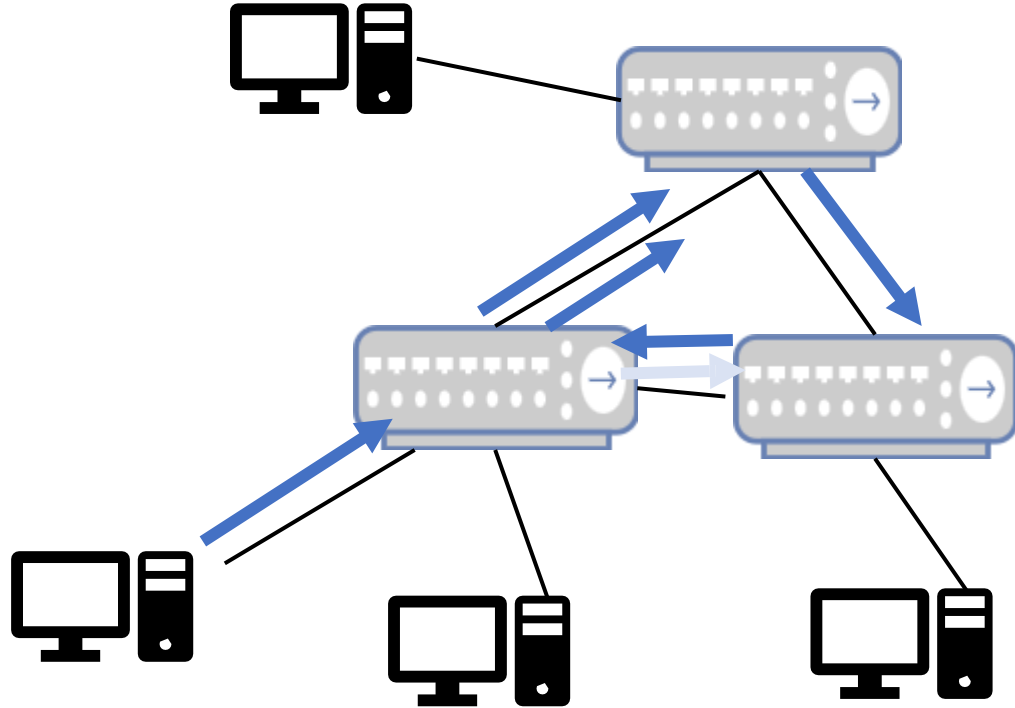
- Recall that there are broadcast MAC addresses
- Recall that we used flooding for learning

Problem: Loops



- Recall that there are broadcast MAC addresses
- Recall that we used flooding for learning

Problem: Loops



- Recall that there are broadcast MAC addresses
- Recall that we used flooding for learning

Solutions to loops

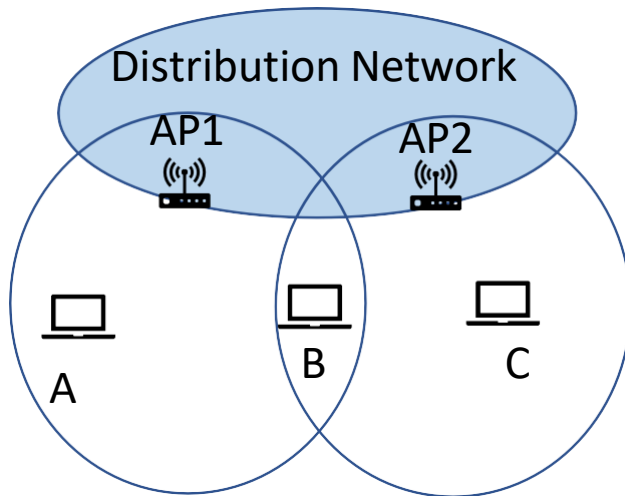
- Could keep track of messages seen
- Could limit the number of hops a given message takes
- Could turn off links that create the loop
(with a protocol like Spanning Tree Protocol)
- Architect network such that looping won't occur
(e.g., topology, or avoid broadcasts and flooding)

In Ethernet LANs, Spanning Tree Protocol is popular, as is architecting the network such that looping won't occur



Wireless LAN

- The special nodes in this case are access points
- Access points commonly connected to a wired network
- All traffic goes via the access points (even if nodes are in range)
- Key challenge here is associating with an access point





University of Colorado **Boulder**